

BENJAMIN PORRECA

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EDUCATION

New Jersey Institute of Technology – Newark, NJ May 2026 B.S Computer Science | GPA:3.300/4.000 – Dean's List

PROFESSIONAL EXPERIENCE

Software Testing and Implementation Intern – June 2024 – August 2024

- Assisted 200+ front-end users with the adoption and navigation of a new tablet-based Commercial Driver's License (CDL) testing software, enhancing user experience and streamlining operations.
- Independently created a testing software to validate CDL test outcomes, reducing upload errors and improving mainframe data reliability.
- Learned and applied mainframe codes and conducted scenario-based tests using Excel matrices to verify software performance.

Vanity: Full-Stack, Role-Based Salon Platform – September 2025 – December 2025

- Designed and implemented a full-stack, role-based e-commerce web application from scratch supporting Customer, Owner, and Admin workflows.
- Collaborated in a Scrum-based Agile environment to deliver 34+ end-to-end, role-based user workflows spanning frontend UI, backend APIs, and persistent data storage.
- Deployed the application to AWS using a CI/CD pipeline, enabling automated builds and continuous production deployment.

PROJECTS

Hand-Tracking Gesture Control

- Developed and executed a real-time hand-tracking system using Python, Mediapipe, and OpenCV.
- Optimized live webcam input to detect and recognize gestures at 30+ FPS for intuitive computer interaction.
- Showcased practical applications of computer vision and machine learning in natural user interfaces.

Custom Falling Sand Engine

- Created a physics-based sandbox in C++ and OpenGL that supports 10+ dynamically interacting particle types, each belonging to one of five different inherited classes.
- Simulated physical properties such as density, friction, and flow for realistic behaviors.
- Demonstrated technical efficiency, scalability, and creativity in real-time graphics programming.

Speech to Input Controller

- Built a real-time speech-to-input Python application using libraries such as Vosk and PyAudio to translate voice commands into keyboard actions.
- Optimized audio processing, command handling, and response timing to achieve low-latency performance suitable for gaming, demonstrating strong skills in real-time systems and practical NLP integration.

LEADERSHIP

External Secretary, Psi Upsilon Fraternity December 2024 – December 2025

- Delivered weekly organizational updates to a network of 100+ alumni and represented the chapter at biweekly IFC meetings.
- Coordinated undergraduate and alumni events, organizing yearly banquets, selecting venues, and managing fundraising initiatives for our philanthropy.

SKILLS & HONORS

Technical: Python, C++, JavaScript, React.js, Flask, MySQL, Git/GitHub, Computer Vision, OpenCV, PyTorch, Tensorflow, Mediapipe, OpenGL, Cellular Automata, Simulation, Robotics, Machine Learning

Coursework: Advanced Data Structures & Algorithms, Cybersecurity, Operating Systems, Internet Applications, Data Science, Software Engineering, Database Systems, Systems Programming, Computer Architecture, Discrete Structures, Linear Algebra, Probability & Statistics

Clubs: NJIT Information & Cybersecurity Club (NICC), NJIT Swim Club